Simple Network Management Protocol (SNMP)

Topology

Initial Configuration Commands

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
line con 0
logging synchronous
```

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```
exit
no ip domain-lookup
int gig 0/1
ip address 192.0.2.100 255.255.255.0
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
end
copy run star
```

INTERNET:

```
enable
conf t
host INTERNET
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 192.0.2.1 255.255.255.0
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
int lo0
ip address 1.1.1.1 255.255.255.255
ip http server
end
copy run star
```

(##This lab uses the same topology and base configuration as the Network Time Protocol(NTP) Lab.)

SNMPv2c Lab Tasks

1. To start, we'll configure SNMPv2c on R1. First, we'll create an SNMP community string named CCNA and assign it read-only permissions.

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2. Next, we will create an SNMP community string named CCNP and assign it read-write permissions.
3. Let's set the device's location to Richmond, KY, and designate Kevin Wallace as the contact person for this device.
4. Let's point to a host at 3.3.3.3, known as the SNMP Manager, set the version to 2c, and use the community string CCNP to send information to the server.
5. Let's enable SNMP traps on R1 by enabling the "snmp-server enable traps" command. Let's see what traps have been enabled by using the "show run | i traps" command.
6. Disable the "snmp-server enable traps ospfv3 errors" trap message on R1.

SNMPv3 Lab Tasks

1. Let's go ahead and start setting up SNMPv3 on R1. Set your engineID to local and give it a number of 0123456789.
2. Let's go ahead and create an SNMP group and name it DEMO_GROUP, and set it for version 3. Next, set our security to do both encryption and authentication.
3. Next, we'll create a user named DEMO_USER who will be added to that group. We'll configure the user to use version 3, set up authentication, choose SHA as the hashing algorithm, and establish the password as "auth-cisco." We'll use PRIV as our authentication parameters, AES as the encryption algorithm with a 128-bit key, and set the privacy password for encryption to "encr-cisco."
4. Run the "show run | i snmp" command, and see if we can find the DEMO_USER user that we created.
5. Run the "show snmp user" command and see if we can see our user now.
6. For our final step, let's associate our user with an SNMP server. We'll set the SNMP server to the host address 4.4.4.4, configure it to version 3, and establish PRIV as the security level for the user DEMO_USER.

Solution for SNMPv2c

Step 1: To start, we'll configure SNMPv2c on R1. First, we'll create an SNMP community string named CCNA and assign it read-only permissions.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#snmp-server community CCNA ro
```

Step 2: Next, we will create an SNMP community string named CCNP and assign it read-write permissions.

```
R1(config)#snmp-server community CCNP rw
```

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Step 3: Let's set the device's location to Richmond, KY, and designate Kevin Wallace as the contact person for this device.

```
R1(config)#snmp-server location Richmond, KY
R1(config)#snmp-server contact Kevin Wallace
```

Step 4: Let's point to a host at 3.3.3.3, known as the SNMP Manager, set the version to 2c, and use the community string CCNP to send information to the server.

```
R1(config)#snmp-server host 3.3.3.3 version ?
  1   Use SNMPv1
  2c  Use SNMPv2c
  3   Use SNMPv3

R1(config)#snmp-server host 3.3.3.3 version 2c CCNP
```

Step 5: Let's enable SNMP traps on R1 by enabling the "snmp-server enable traps" command. Let's see what traps have been enabled by using the "show run | i traps" command.

```
R1(config)#snmp-server enable traps
R1(config)#exit
R1#show run | i traps
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps pfr
snmp-server enable traps flowmon
snmp-server enable traps tty
snmp-server enable traps eigrp
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps diameter
snmp-server enable traps vrrpv3
snmp-server enable traps auth-framework sec-violation auth-fail
snmp-server enable traps trustsec-sxp conn-srcaddr-err msg-parse-err conn-config-err
binding-err conn-up conn-down binding-expn-fail oper-nodeid-change binding-conflict
snmp-server enable traps aaa_server
snmp-server enable traps atm subif
snmp-server enable traps bfd
snmp-server enable traps bgp cbgp2
```

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```
snmp-server enable traps bstun
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change
inconsistency
snmp-server enable traps memory bufferpeak
snmp-server enable traps cnpd
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps dlsw
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps entity-ext
snmp-server enable traps entity
snmp-server enable traps fru-ctrl
snmp-server enable traps resource-policy
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps hsrp
snmp-server enable traps ipmobile
snmp-server enable traps ipmulticast
snmp-server enable traps isis
snmp-server enable traps mempool
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps pppoe
snmp-server enable traps cpu threshold
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps rsvp
snmp-server enable traps ipsla
snmp-server enable traps stun
snmp-server enable traps syslog
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps pw vc
snmp-server enable traps waas
snmp-server enable traps event-manager
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pki
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
```

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```
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps rf
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps mpls vpn
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
```

(##We have to enable trap messages to let R1 know what events would cause R1 to communicate with the server at 3.3.3.3.)

Step 6: Disable the "snmp-server enable traps ospfv3 errors" trap message on R1.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no snmp-server enable traps ospfv3 errors
```

(##We can remove traps if they aren't compatible with what we are doing on our network.)

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Solution for SNMPv3

Step 1: Let's go ahead and start setting up SNMPv3 on R1. Set your engineID to local and give it a number of 0123456789.

```
R1(config)#snmp-server engineID local 0123456789

(##The engine-ID is a 10 digit hexadecimal number that
identifies the SNMP instance on this device.  It is not
required, but it is recommended for documentation.)
```

Step 2: Let's go ahead and create an SNMP group and name it DEMO_GROUP, and set it for version 3. Next, set our security to do both encryption and authentication.

```
R1(config)#snmp-server group DEMO_GROUP ?
  v1    group using the v1 security model
  v2c   group using the v2c security model
  v3    group using the User Security Model (SNMPv3)

R1(config)#snmp-server group DEMO_GROUP v3 ?
  auth     group using the authNoPriv Security Level
  noauth   group using the noAuthNoPriv Security Level
  priv     group using SNMPv3 authPriv security level

R1(config)#snmp-server group DEMO_GROUP v3 priv
```

Step 3: Next, we'll create a user named DEMO_USER who will be added to that group. We'll configure the user to use version 3, set up authentication, choose SHA as the hashing algorithm, and establish the password as "auth-cisco." We'll use PRIV as our authentication parameters, AES as the encryption algorithm with a 128-bit key, and set the privacy password for encryption to "encr-cisco."

```
R1(config)#snmp-server user DEMO_USER DEMO_GROUP v3 ?
  access     specify an access-list associated with this group
  auth       authentication parameters for the user
  encrypted  specifying passwords as MD5 or SHA digests
  <cr>       <cr>

R1(config)#snmp-server user DEMO_USER DEMO_GROUP v3 auth ?
  md5  Use HMAC MD5 algorithm for authentication
  sha  Use HMAC SHA algorithm for authentication

R1(config)#snmp-server user DEMO_USER DEMO_GROUP v3 auth sha ?
  WORD  authentication password for user
```

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```
R1(config)#snmp-server user DEMO_USER DEMO_GROUP v3 auth sha auth-cisco ?
  access  specify an access-list associated with this group
  priv    encryption parameters for the user
  <cr>    <cr>

R1(config)#snmp-server user DEMO_USER DEMO_GROUP v3 auth sha auth-cisco priv
?
  3des  Use 168 bit 3DES algorithm for encryption
  aes   Use AES algorithm for encryption
  des   Use 56 bit DES algorithm for encryption

R1(config)#$ user DEMO_USER DEMO_GROUP v3 auth sha auth-cisco priv aes ?
  128  Use 128 bit AES algorithm for encryption
  192  Use 192 bit AES algorithm for encryption
  256  Use 256 bit AES algorithm for encryption

R1(config)#$ user DEMO_USER DEMO_GROUP v3 auth sha auth-cisco priv aes 128 ?
  WORD  privacy password for user

R1(config)#$_USER DEMO_GROUP v3 auth sha auth-cisco priv aes 128 encr-cisco
R1(config)#end
```

(##We are choosing sha as our hashing algorithm to take a fingerprint of the password that we are entering so that it can't be reverse engineered. Sha is more secure than md5.)
(##AES is the flagship encryption algorithm today. The more bits we use, the more secure it is.)

Step 4: Run the "show run | i snmp" command, and see if we can find the DEMO_USER user that we created.

```
R1#show run | i snmp
mmi snmp-timeout 180
snmp-server engineID local 0123456789
snmp-server group DEMO_GROUP v3 priv
snmp-server community CCNA RO
snmp-server community CCNP RW
snmp-server location Richmond, KY
snmp-server contact Kevin Wallace
snmp-server chassis-id
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps pfr
snmp-server enable traps flowmon
snmp-server enable traps tty
snmp-server enable traps eigrp
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
```

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```
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps diameter
snmp-server enable traps vrrpv3
snmp-server enable traps auth-framework sec-violation auth-fail
snmp-server enable traps trustsec-sxp conn-srcaddr-err msg-parse-err conn-config-err
binding-err conn-up conn-down binding-expn-fail oper-nodeid-change binding-conflict
snmp-server enable traps aaa_server
snmp-server enable traps atm subif
snmp-server enable traps bfd
snmp-server enable traps bgp cbgp2
snmp-server enable traps bstun
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change
inconsistency
snmp-server enable traps memory bufferpeak
snmp-server enable traps cnpd
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps dlsw
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
```

(##As you can see, both SNMPv2c and SNMPv3 configurations are present in our running config.)
(##You'll notice that we can't see our SNMP user. By requirement of SNMPv3, our username can't be in the running config.)

Step 5: Run the "show snmp user" command and see if we can see our user now.

```
R1#show snmp user

User name: DEMO_USER
Engine ID: 0123456789
storage-type: nonvolatile        active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: DEMO_GROUP
```

(##Here, we can now see our SNMPv3 user.)

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Step 6: For our final step, let's associate our user with an SNMP server. We'll set the SNMP server to the host address 4.4.4.4, configure it to version 3, and establish PRIV as the security level for the user DEMO_USER.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#snmp-server host 4.4.4.4 version ?
  1   Use SNMPv1
  2c  Use SNMPv2c
  3   Use SNMPv3

R1(config)#snmp-server host 4.4.4.4 version 3 ?
  auth    Use the SNMPv3 authNoPriv Security Level
  noauth  Use the SNMPv3 noAuthNoPriv Security Level
  priv    Use the SNMPv3 authPriv Security Level

R1(config)#snmp-server host 4.4.4.4 version 3 priv ?
  WORD  SNMPv1/v2c community string or SNMPv3 user name

R1(config)#snmp-server host 4.4.4.4 version 3 priv DEMO_USER
```

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